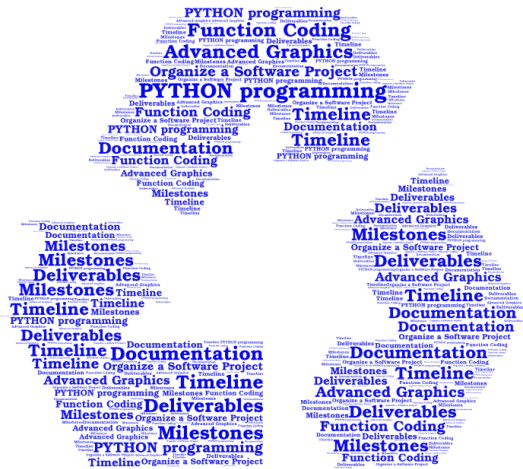




Project Work: Initiation of the project 'Scooter'

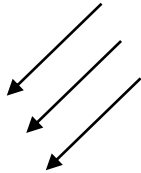
Lecture: Modelling an Simulation

This is a compulsory elective module
of
**Technology of Circular
Economy**
(Master of Engineering)

Dr. Uwe Reimer
Senior Scientist

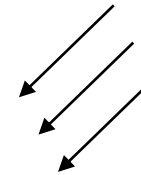
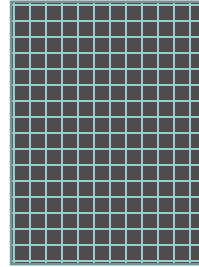
Design a PV system which collects energy from the sun and stores it in a battery. An electric scooter shall be charged with this energy for commuting to work (daily drive cycle).

Results shall be presented for the city of Emden and second location in a different country.



Singe PV module

- Peak Power 500 W
- Size 2 m x 1 m



What do you need to calculate the power output?



The five project management phases.

- 1) Initiation
- 2) Planning
- 3) Execution
- 4) Monitoring and control
- 5) Closure

How do you start?

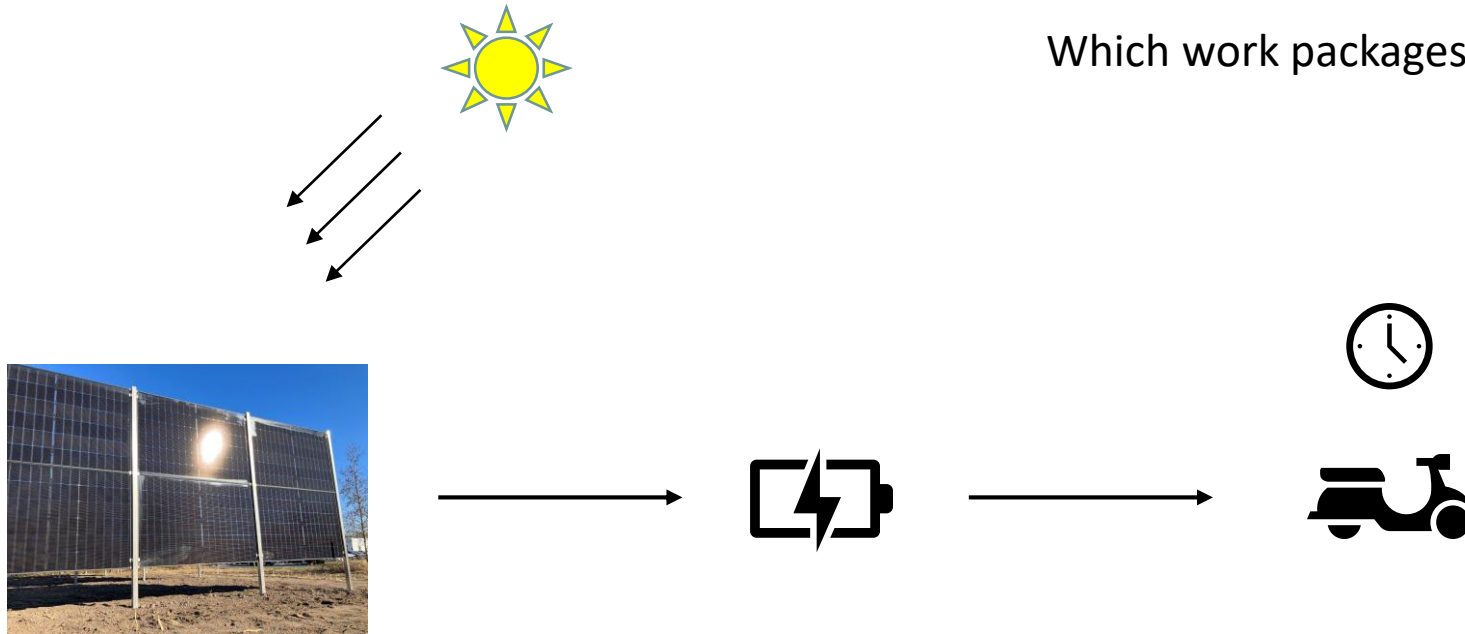
These steps should be followed.

- 1) Breakdown of the overall problem
- 2) Define WPs
- 3) Define deliverables
- 4) Create time line and Gantt chart
- 5) Define milestones
- 6) Write down software requirements
- 7) Create Kanban board

Break down of problem

Analyze work. Break down in useful work packages.

Which work packages can we identify here?



Identify work packages

- location for PV WP 1
- charging / battery storage WP 2
- sun data WP 1
- data for scooter WP 2
- software WP 3

- cost calculation WP 4

WP 1: Scooter

- Drive cycle and energy demand
- Result = energy demand per day / safety margin?

WP 2: Solar panel

- Technical specifications
- How to generate power output for one day?
- How to generate power output for one year?
- Decision on size

WP 3: Battery

- Decision on size
(We use an idealized battery, only consider the needed storage capacity.)
- Needs power demand from WP 1 and power output from WP 2

Identify work packages

WP 1: Scooter

- Drive cycle and energy demand
- Result = energy demand per day / safety margin?

WP 2: Solar panel

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- How to generate power output for one year?
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WP 4: System Software Design

- How should the components communicate?
- Time steps within the system
- Which input is needed?
- What should be the output?

- 1) Breakdown of the overall problem
- 2) Define WPs
- 3) Define deliverables
- 4) Create time line and Gantt chart
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WP 1: Scooter

- Drive cycle and energy demand
- Result = energy demand per day / safety margin?



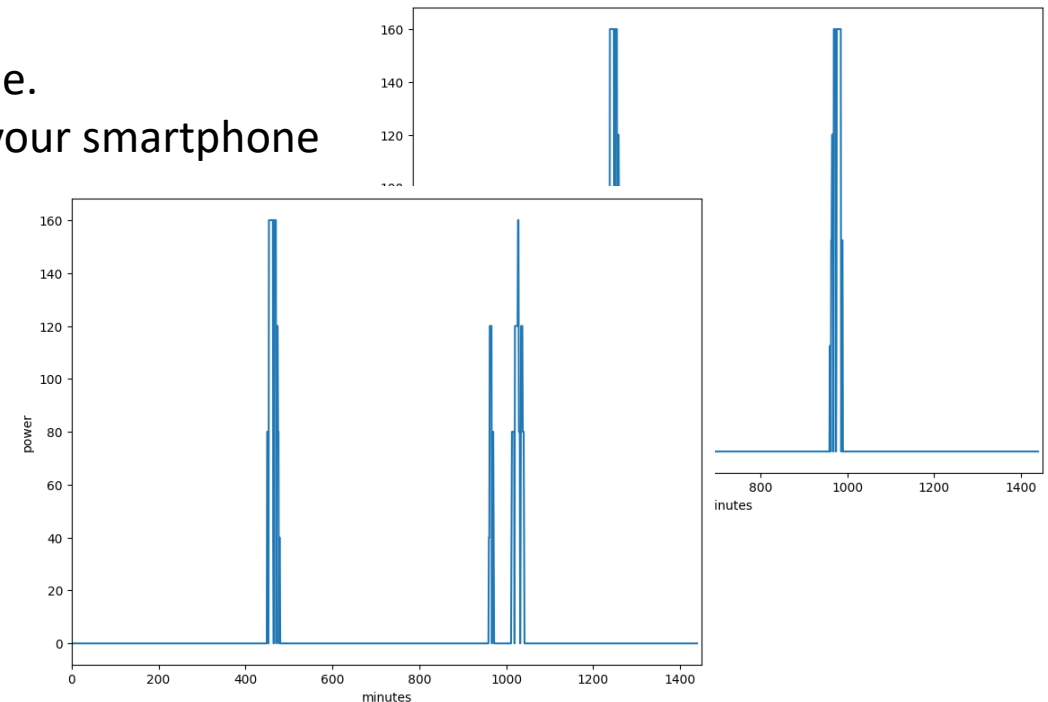
We will use a simulated drive cycle.
Imagine, this was recorded with your smartphone
while driving (resolution 1 min).

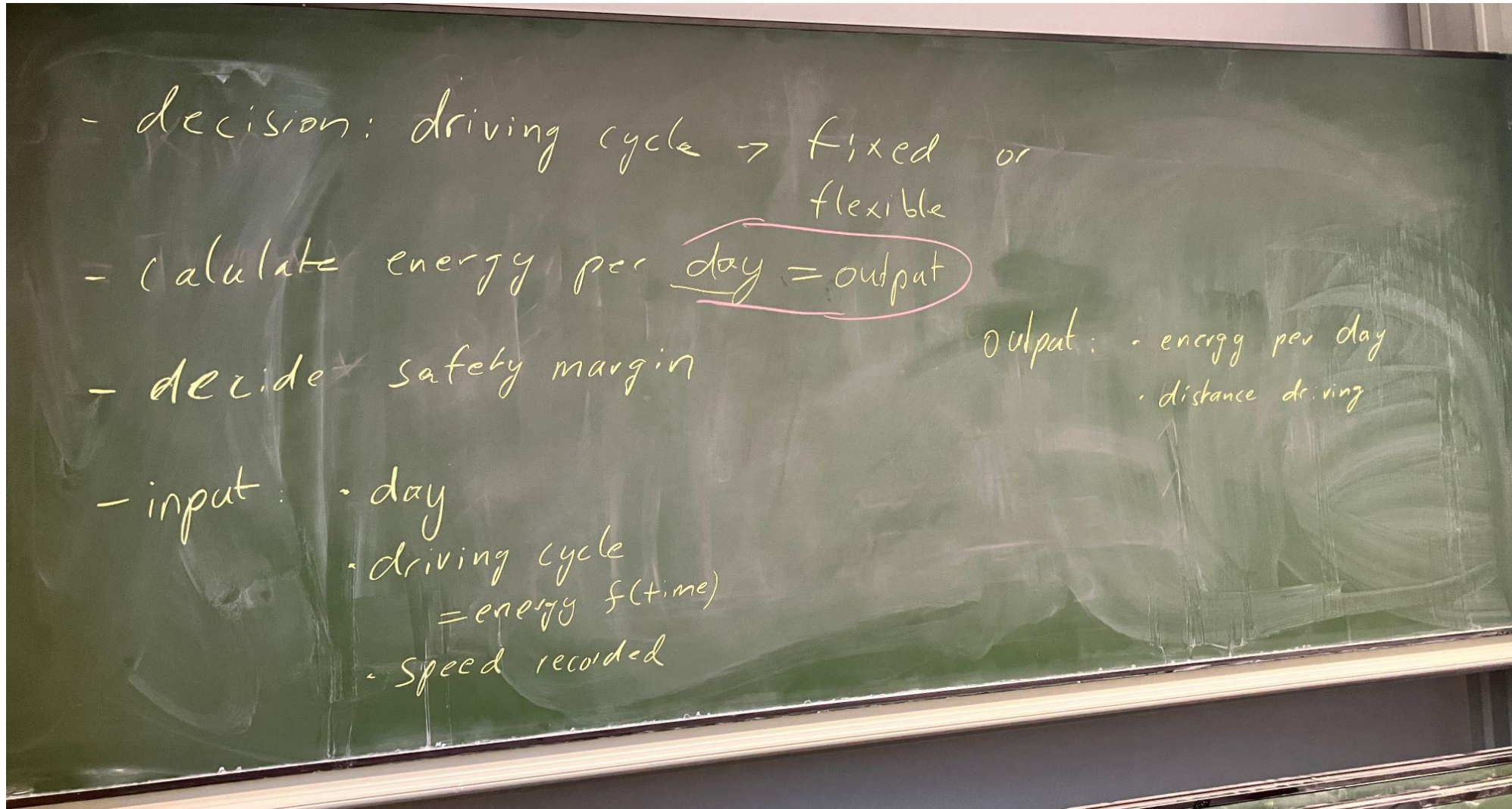
What is input?

What is output?

What needs to be calculated?

Which assumptions do you make?





WP 1: Scooter

- Drive cycle and energy demand
- Result = energy demand per day / safety margin?

What is input?

What is output?

What needs to be calculated?

Which assumptions do you make?

Work package 1 / WP 1: Scooter

Input:

1. Day,
2. Driving Cycle
3. Energy $f(\text{time})$
4. Speed recorded

Output:

1. Energy per day
2. Distance driving

Deliverable D1: Definition of scooter energy demand per day

Work package 1 / WP 1

- Input
 - day
 - driving cycle energy $f(\text{time})$
 - Speed recorded
- Output
 - energy per day
 - distance driving

**Deliverable D1: Definition of scooter energy demand per day.
(Scooter model ready)**

WP 2: Solar panel

- Technical specifications
- How to generate power output for one day?
- How to generate power output for one year?
- Decision on size

What do you need to calculate the power output?

What is input?

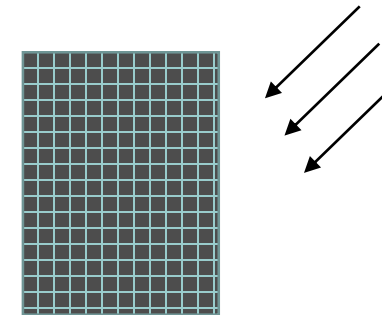
What is output?

What needs to be calculated?

Which assumptions do you make?

Singe PV module

- Peak Power 500 W
- Size 2 m x 1 m



WP 2: Solar panels

WP 2: Solar panel

- Technical specifications
- How to generate power output for one day?
- How to generate power output for one year?
- Decision on size

What is input?

What is output?

What needs to be calculated?

Which assumptions do you make?

Work package 2 / WP 2: Solar

Input:

1. Location of PV
2. Size of the PV , Number of PV
3. Angle
4. Time duration per day
5. Maximum capacity
6. Sun data

Output:

1. generation of energy per day

Assumptions:

1. Sunlight per day
2. Energy losses

Deliverable D2: Generation of energy demand for the PV per day

Work package 2 / WP 2

- Input:

-(Output WP1)

-PV Module size

-Power generation per module

-meteorological data

- Output:

-Number of PV Modules

-power output

**Deliverable D2: Calculate a reference for the daily power output for one year.
(Solar model ready)**

WP 3: Battery

- Decision on size
(We use an idealized battery, only consider the needed storage capacity.)
- Needs power demand from WP 1 and power output from WP 2

What is input?

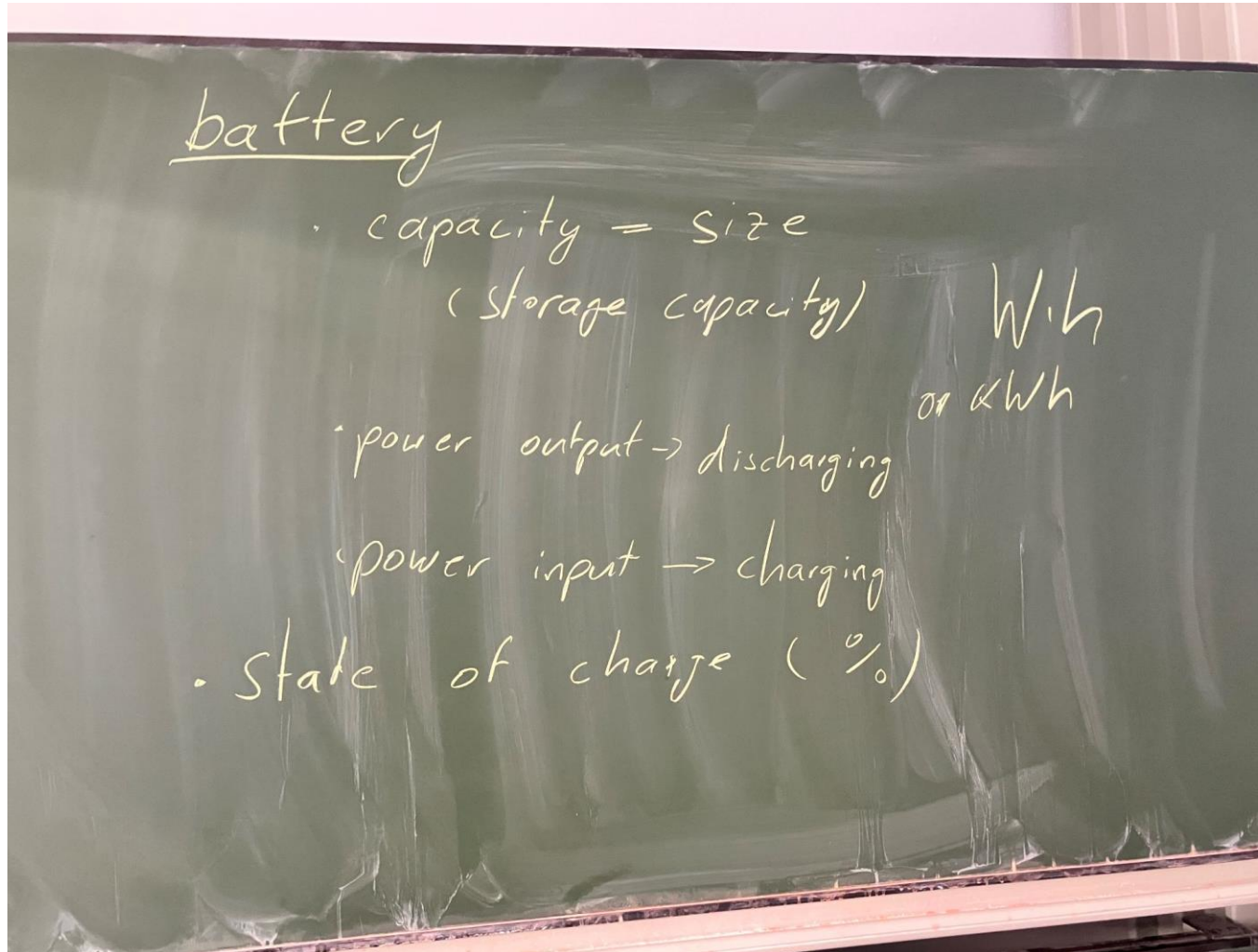
What is output?

What needs to be calculated?

Which assumptions do you make?

(How many days power? Safety margins? There are also cloudy days.)





WP 3: Battery

- Decision on size
(We use an idealized battery, only consider the needed storage capacity.)
- Needs power demand from WP 1 and power output from WP 2

Work package 3 / WP 3: Battery

Input:

1. Battery capacity
2. Solar generated energy per day (Charging)

Output:

1. Power distribution to the scooter (Discharging)

Assumptions:

1. Battery temperature
2. Battery life

Deliverable D3: State of charge (%) (Battery storage)

Work package 3 / WP 3

• Input:

- energy demand
- energy generation
- Design and safety constraints

• Output:

- storage capacity
- Performance
- Battery management system

• Assumptions:

- Battery efficiency(90-95%)
- constant sunlight (4 hrs per day)
- Battery life

• Deliverables:

- Specification sheet

What is input?

What is output?

What needs to be calculated?

Which assumptions do you make?

(How many days power? Safety margins? There are also cloudy days.)

Deliverable D3: Definition of storage capacity for battery.

(Battery model ready)

WP 4: System Software Design

- How should the components communicate?
- Time steps within the system
- Which input is needed?
- What should be the output?

Possible software system configurations:

The system has 3 components:

- PV panel
- battery
- scooter

WP 4: System Software Design

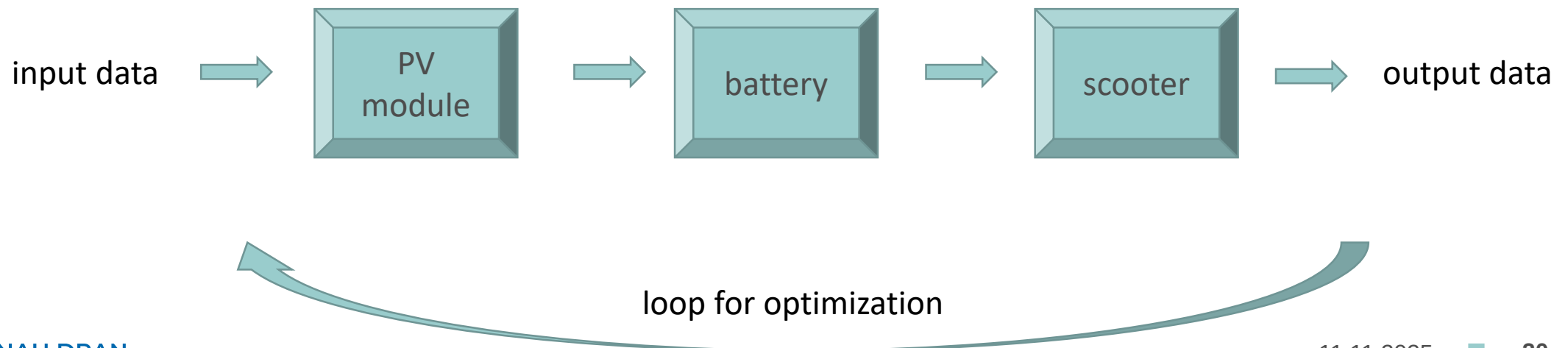
- How should the components communicate?
- Time steps within the system
- Which input is needed?
- What should be the output?

The system has 3 components:

- PV panel
- battery
- scooter

Possible software system configurations:

Linear configuration (needs interface between all components)

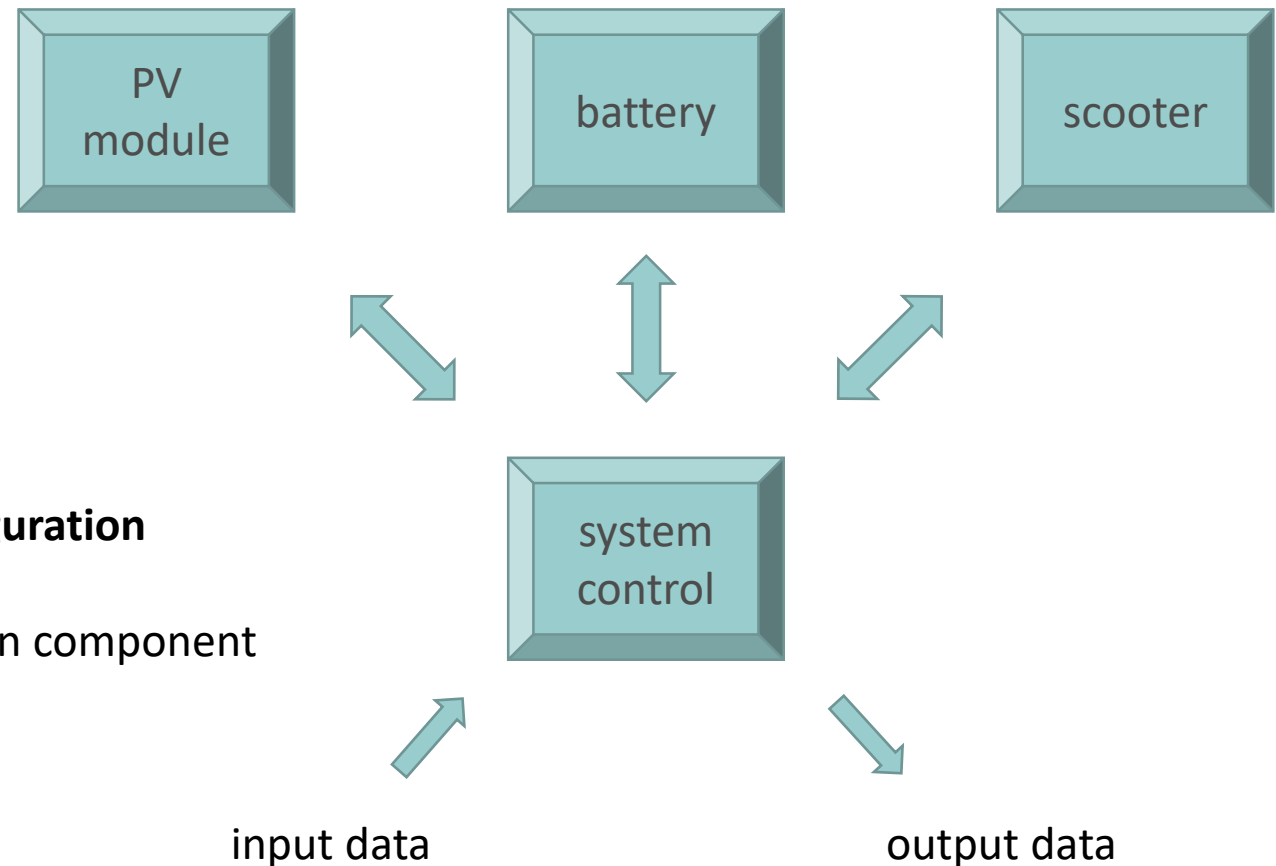


WP 4: System Software Design

- How should the components communicate?
- Time steps within the system
- Which input is needed?
- What should be the output?

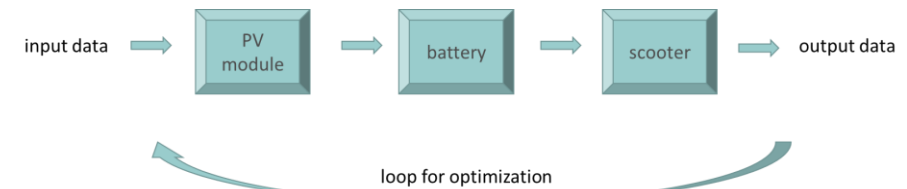
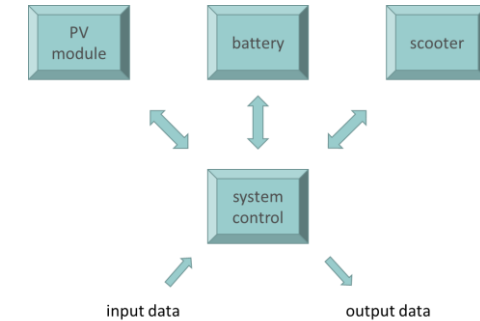
System controller configuration

(needs interface between component and controller)



WP 4: System Software Design

- How should the components communicate?
- Time steps within the system
- Which input is needed?
- What should be the output?



Deliverable D4: Definition of software system.
(System model ready)

- 1) Breakdown of the overall problem 
- 2) Define WPs 
- 3) Define deliverables 
- 4) Create time line and Gantt chart
- 5) Define milestones
- 6) Write down software requirements
- 7) Create Kanban board

Including today, there are 6 weeks until christmas.

Create a Gantt chart to finish the project 'Scooter' until christmas.

Including today, there are 6 weeks until christmas.

Create a Gantt chart to finish the project 'Scooter' until christmas.

		week						
Work package		1	2	3	4	5	6	
	Planning							
WP 1	Scooter							
WP 2	Solar panel							
WP 3	Battery							
WP 4	System Design							
WP 5	Calculate results							
	Write report							

Now, create useful milestones.

Including today, there are 6 weeks until christmas.

Example:

		week						
Work package		1	2	3	4	5	6	
	Planning	*	M 1					
WP 1	Scooter							
WP 2	Solar panel							
WP 3	Battery				*	M 2		
WP 4	System Design							
WP 5	Calculate results						*	M 3
	Write report							
List of Deliverables		List of Milestones						
	D 1		M 1	Plan ready				
	D 2		M 2	All models ready				
	D 3		M 3	Results ready				
	D 4							
	D 5							

- 1) Breakdown of the overall problem 
- 2) Define WPs 
- 3) Define deliverables 
- 4) Create time line and Gantt chart 
- 5) Define milestones 
- 6) Write down software requirements
- 7) Create Kanban board

For practical use: the requirements should contain

- Purpose (see project description)
- Overall description (see work package definition)
- Specific requirements (This is what we need to do now!)

We have four software developing work packages.

WP 1 Scooter

WP 2 Solar panel

WP 3 Battery

WP 4 System Design

For each of these WP we need to go through the MoSCoW method. Again, this will be group work. Please go to the Moodle system and follow the link for your group.

We will take 10 min for WP 1 and then compare the results.

After the end of the lecture there will be a compilation of results in the lecture notes.

M	Must have
S	Should have
C	Could have
W	Won't have

Work package 1 / WP 1: Scooter

Input:

1. Day : **M**
2. Driving Cycle : **M**
3. Energy $f(\text{time})$: **S**
4. Speed recorded : **C**

Output:

1. Energy per day : **M**
2. Distance driving: **C / W**

Deliverable D1: Definition of scooter energy demand per day

.

Work package 2 / WP 2: Solar

Input:

1. Location of PV: **S**
2. Size of the PV , Number of PV: **M**
3. Angle: **S**
4. Time duration per day: **C**
5. Maximum capacity: **M**
6. Sun data: **S**

Output:

1. generation of energy per day: **M**

Assumptions:

1. Sunlight per day: **C / W**
2. Energy losses: **W**

Deliverable D2: Generation of energy demand for the PV per day

Work package 3 / WP 3: Battery

Input:

1. Battery capacity: **M**
2. Solar generated energy per day (Charging): **S**

Output:

1. Power distribution to the scooter (Discharging): **C**

Assumptions:

1. Battery temperature: **W**
2. Battery life: **C**

Deliverable D3: State of charge (%) (Battery storage)

.

Work package 4 / WP 4: System Software Design

Input:

1. Generation of energy per day: **M**
2. Distance driving: **C / W**
3. Power distribution to the scooter (Discharging): **C**
4. State of charge (%): **C / W**
5. Speed: **C / W**

Output:

1. Charging and discharging rate : **C**
2. Distance driving: **C / W**

Assumptions:

1. Linear / System Controller : **M**
2. Energy losses: **W**

Deliverable D4: Configred system

- 1) Breakdown of the overall problem 
- 2) Define WPs 
- 3) Define deliverables 
- 4) Create time line and Gantt chart 
- 5) Define milestones 
- 6) Write down software requirements 
- 7) Create Kanban board